Do Now · retrieval

Lesson 138 · SQL SELECT, Filtering, Sorting and JOIN

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What table is created to remove an EmployeeID→DeptID→DeptName transitive dependency?

Q2

When does a linear search stop?

Q3

Which is removed first from a FIFO queue?

Q4

What does ROM stand for?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Choose projection — SELECT chooses columns from a named relation.

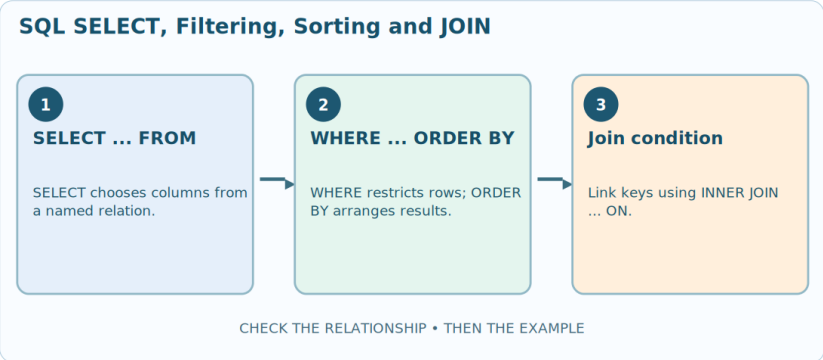
Core idea 2

Filter and order — WHERE restricts rows; ORDER BY arranges results.

Core idea 3

Join related rows — Link keys using INNER JOIN ... ON.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

SELECT — Returns specified columns/expressions.

WHERE — Filters rows according to a condition.

ORDER BY — Sorts returned rows by chosen column/expression.

JOIN — Combines rows from related tables using a join condition.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Orders(OrderID,CustomerID,Total) has totals 25,75,120. Write SQL listing orders over 50 with largest first.

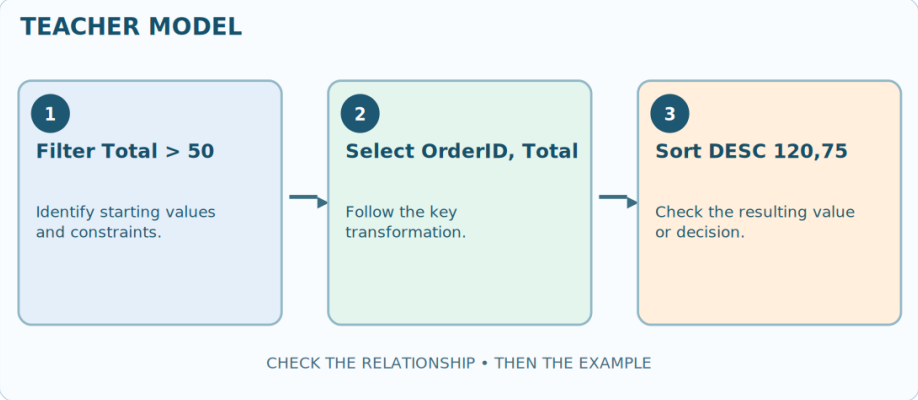
Follow the worked method

1. Filter Total > 50 — Find the given data and name the required result.

2. Select OrderID, Total — Explain why this intermediate stage is necessary.

3. Sort DESC 120,75 — SELECT OrderID, Total FROM Orders WHERE Total > 50 ORDER BY Total DESC; returns orders with 120 then 75, excluding 25.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Join on CustomerID
Find the given data and name the required result.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Write SQL listing Title and Price from Book where Price<10, sorted ascending by Price. [6]

Question 2

Explain the role of the ON clause in an INNER JOIN between Borrower and Loan. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Write an INNER JOIN query listing Student.Name and Course.Title using Student, Enrolment and Course. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

ORDER BY always removes duplicate result rows.

Use key vocabulary from memory

1. Explain SELECT in your own words.

2. Explain WHERE in your own words.

3. Explain ORDER BY in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.